



Modelling Financial Markets

Chapter 4: Portfolio Optimisation



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ DSR: Deflated Sharpe Ratio
- ▣ ERC: Equal Risk Contribution portfolio
- ▣ FDR: False Discovery Rate
- ▣ FWER: Family-Wise Error Rate
- ▣ GLS: Generalised Least Squares
- ▣ GMV: Global Minimum-Variance portfolio
- ▣ GMV-LO: Global Minimum-Variance portfolio, Long-Only
- ▣ GMV-LW: Global Minimum-Variance portfolio with Ledoit-Wolf covariance
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HRP: Hierarchical Risk Parity
- ▣ HYG: iShares iBoxx High Yield Corporate Bond ETF
- ▣ IEF: iShares 7–10 Year Treasury Bond ETF
- ▣ LO: Long-Only, no short sales
- ▣ LQD: iShares iBoxx Investment Grade Corporate Bond ETF
- ▣ ML: Machine Learning
- ▣ MV: Mean–Variance tangency portfolio estimated from sample moments
- ▣ MV-LO: Mean–Variance tangency portfolio, Long-Only
- ▣ NYSE: New York Stock Exchange
- ▣ OLS: Ordinary Least Squares
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ OOS: Out-Of-Sample

Acronyms used in this material (2/2)

- POET: Principal Orthogonal compleMent Thresholding
- US: United States

Chapter Route and Outcomes

The question

- A **portfolio** is a weight vector $w = (w_1, \dots, w_N)^\top$ with $w^\top \mathbf{1} = 1$; optimisation chooses w to maximise a criterion under constraints
- The optimum is easy to write with known means and covariances, hard to estimate: the optimiser loads on the largest estimation errors
 - ▶ 16 assets: $16 + 136 = 152$ parameters, estimated from 60 monthly observations of a 16-dimensional return vector
- Can any estimated rule beat equal weights out of sample, after costs?

Route

- Part I: mean–variance theory, estimation risk, covariance shrinkage, Black–Litterman
- Part II: risk parity and HRP, out-of-sample tests, BVB portfolios, a case study, one pass of the AI-assisted research loop
- Technical detail sits in the [Appendix](#); textbook: [Franke, Härdle & Hafner \(2019\)](#), Part I

Data, Assets and the 1/N Benchmark

Universe	Assets	Returns	Currency
Sectors	9 US sector ETFs: XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY	1999/01 – 2026/07	USD
Multi-asset	SPY, EFA, EEM (equities); TLT, IEF (Treasuries); LQD, HYG (credit); GLD (gold)	2007/05 – 2026/07	USD
Combined	9 sectors + the 7 multi-asset ETFs other than SPY (16 ETFs)	2007/05 – 2026/07	USD
Romania	8 BVB blue chips: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL; BET, BET-TR	2014/10 – 2026/08	RON

- Monthly simple returns from month-end prices on common trading days; ETFs and stocks with dividends reinvested; BET is a price index, BET-TR reinvests dividends
- Excess returns over the one-month Treasury bill ([Kenneth French Data Library](#)); BVB: $R_f = 0$
- Sectors: XLB materials, XLE energy, XLF financials, XLI industrials, XLK technology, XLP staples, XLU utilities, XLV health care, XLY discretionary
- Multi-asset: SPY (S&P 500), EFA (developed-market equities outside the US and Canada), EEM (emerging-market equities), TLT (US Treasuries, 20+ years), IEF (US Treasuries, 7–10 years), LQD (investment-grade corporate bonds), HYG (high-yield corporate bonds), GLD (gold)
- BVB: TLV Banca Transilvania, SNP OMV Petrom, BRD BRD–Groupe Société Générale, TGN Transgaz, SNG Romgaz, SNN Nuclearelectrica, EL Electrica, TEL Transelectrica
- 1/N: equal weights $w_i = 1/N$, rebalanced every month; no estimation at all, the benchmark of every comparison

Notation

- Simple return $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$; excess return $R_{i,t}^e = R_{i,t} - R_{f,t}$; portfolio mean $w^\top \mu$, variance $w^\top \Sigma w$
- **Sharpe ratio** ([Sharpe \(1966\)](#)): $SR_p = \mathbb{E}[R_p^e]/\sigma(R_p^e)$; monthly figures annualised with $\times 12$ (mean) and $\times \sqrt{12}$ (volatility, Sharpe)
- **Maximum drawdown**: $\min_t(W_t/\max_{s \leq t} W_s - 1)$, the largest fall of wealth W_t from a previous peak
- **Out of sample** (OOS): the weights held in month t use only data up to month $t - 1$
- Rules of this chapter: MV (tangency from sample moments), GMV (global minimum variance), their long-only versions MV-LO and GMV-LO, GMV-LW (GMV with shrinkage), ERC, HRP and 1/N

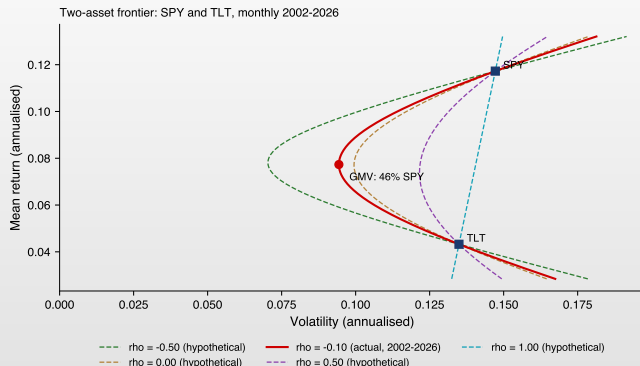
The Markowitz Problem

- ▣ **Markowitz (1952)**: investors care about the mean and the variance of portfolio returns
- ▣ **Minimum-variance frontier**: minimum variance for every target mean m ; its upper branch, $m \geq \mu_{\text{GMV}}$ (the mean of the GMV portfolio), is the **efficient frontier**

$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu = m, \quad w^\top \mathbf{1} = 1$$

- ▣ Utility form, risk aversion $\gamma > 0$: $\max_w w^\top \mu - \frac{\gamma}{2} w^\top \Sigma w$ s.t. $w^\top \mathbf{1} = 1$; as γ varies it traces exactly the efficient branch
- ▣ Weights may be negative: a **short sale** (selling a borrowed asset)

Two Assets: How Correlation Diversifies



- Two assets, correlation ρ :

$$\sigma_p^2 = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2$$
- Minimised at $w^* = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$
- SPY and TLT, monthly 2002–2026:
 $\rho = -0.10$, $w^* = 46\%$ SPY, volatility 9.4% against 14.7% and 13.5%
- The lower ρ , the more the curve bends to the left; at $\rho = 1$ there is no diversification
- Time-varying correlation: [Chapter 6](#)

GMV and Tangency Portfolios for N Assets

- GMV: $\min_w w^\top \Sigma w$ s.t. $w^\top 1 = 1$; Lagrangian $\mathcal{L} = w^\top \Sigma w - \lambda(w^\top 1 - 1)$, Σ positive definite
 - First-order condition $2\Sigma w = \lambda 1$, so $w = \frac{\lambda}{2} \Sigma^{-1} 1$; the constraint gives $\frac{\lambda}{2} = 1/A$, $A = 1^\top \Sigma^{-1} 1$
- Tangency: maximise $w^\top \mu / \sqrt{w^\top \Sigma w}$, μ the expected **excess** returns; the condition $\mu \propto \Sigma w$ gives $w \propto \Sigma^{-1} \mu$

	GMV	Tangency (MV)
Weights	$w_{\text{GMV}} = \Sigma^{-1} 1 / A$	$w_{\text{MV}} = \Sigma^{-1} \mu / B$, $B = 1^\top \Sigma^{-1} \mu$
Result	variance $\sigma_{\text{GMV}}^2 = 1/A$	squared Sharpe $\theta^2 = \mu^\top \Sigma^{-1} \mu$
Needs	Σ only	μ and Σ ; Sharpe $\text{sign}(B) \theta$

- The normalised $\Sigma^{-1} \mu / B$ is the maximum only if $B > 0$; $B < 0$ gives the minimum-Sharpe portfolio, $B = 0$ no fully invested solution

Separation and the Frontier in Closed Form



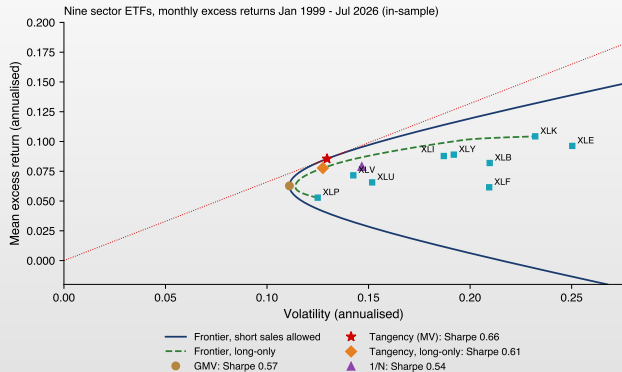
James Tobin, Nobel Prize in
Economics 1981

Photo: unknown author (1962); public
domain; Wikimedia Commons

- **Tobin (1958)**: add a riskless asset paying R_f ; every investor holds the **same** risky portfolio, the tangency portfolio, mixed with cash or borrowing
 - ▶ Risk aversion changes only the mix along the capital allocation line: **two-fund separation**, the bridge to the CAPM ([Chapter 3](#))
- **Merton (1972)**: with short sales, the frontier is a hyperbola; Σ positive definite, μ not proportional to 1
 - ▶ $A = 1^\top \Sigma^{-1} 1$, $B = 1^\top \Sigma^{-1} \mu$, $C = \mu^\top \Sigma^{-1} \mu$, $D = AC - B^2 > 0$

$$\sigma^2(m) = \frac{Am^2 - 2Bm + C}{D}$$
- The GMV sits at $m = B/A$, $\sigma^2 = 1/A$; every frontier portfolio mixes any two distinct frontier portfolios, e.g. GMV and tangency (if $B \neq 0$)

The Frontier of US Sectors



- Monthly excess returns, January 1999 – July 2026 ($T = 331$); all parameters estimated on the full sample: an **in-sample** picture
- Tangency: Sharpe 0.66; GMV: 0.57 (volatility 11.1%); $1/N$: 0.54
- Tangency weights: -0.37 financials, -0.21 materials, $+0.37$ health care: short positions
- Long-only tangency: Sharpe 0.61; materials, financials and industrials get exactly zero, a **corner solution**
- The gain over $1/N$ is small even with perfect hindsight

Constraints

- ▣ **Long-only (LO)**: $w_i \geq 0$; no closed form, solved by quadratic programming
- ▣ Position limits: $w_i \leq 0.2$; sector or asset-class limits: $\sum_{i \in G} w_i \leq c_G$
- ▣ Leverage limit: $\sum_i |w_i| \leq L$
- ▣ **Frost & Savarino (1988)**: constraining weights improved out-of-sample performance; why this works is a shrinkage result (next section)
- ▣ MV-LO and GMV-LO are the long-only versions of MV and GMV

Means Are Hard to Estimate: Merton's Argument

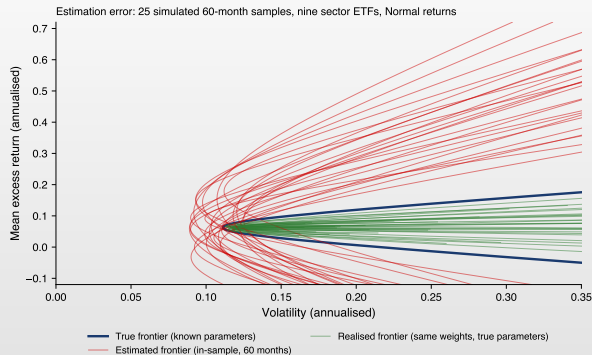


Robert C. Merton, Nobel Prize in
Economics 1997

Photo: Digarnick (2006); public
domain; Wikimedia Commons

- Merton (1980): Y years sampled h times a year, i.i.d. Normal returns (a diffusion) with annual mean μ and variance σ^2
 - ▶ $\text{Var}(\hat{\mu}) = \sigma^2/Y$: independent of h ; only the **span** of the sample helps
 - ▶ $\text{Var}(\hat{\sigma}^2) \approx 2\sigma^4/(hY) \rightarrow 0$ as $h \rightarrow \infty$ (Normal; in general $(\kappa - 1)\sigma^4/(hY)$, κ the kurtosis): variances sharpen with frequency
- SPY, $Y = 24$, $\sigma = 14.7\%$: standard error 3.0%, 95% interval [5.8%, 17.6%] for a mean of 11.7%
- A standard error of 1% needs $(0.147/0.01)^2 \approx 217$ years
- Chopra & Ziemba (1993): errors in means cost about ten times more than errors in variances; rules without means (GMV, risk parity) are more robust

Error Maximisation: Estimated versus Realised Frontiers



- **Michaud (1989):** optimisers are **estimation-error maximisers**; **Best & Grauer (1991):** a tiny change in one mean moves many weights
- True parameters = sector moments 1999–2026; draw 60 Normal months, optimise, evaluate with the true parameters
- Red: estimated frontiers, too optimistic; green: the same portfolios under the true parameters, all below the true frontier
- 2000 simulated tangency portfolios: median estimated maximum Sharpe 1.56, median true Sharpe 0.30, true optimum 0.66
- 96% have a true Sharpe below $1/N$ (0.54), 7.5% below zero; in 11% of samples $1^\top \hat{\Sigma}^{-1} \hat{\mu} < 0$

The Plug-in Sharpe Ratio Is Biased

- i.i.d. $N(\mu, \Sigma)$ returns, maximum-likelihood $\hat{\mu}$, $\hat{\Sigma}$, $\hat{\theta}^2 = \hat{\mu}^\top \hat{\Sigma}^{-1} \hat{\mu}$ (Kan & Zhou (2007); Wishart moments in the [Appendix](#))

$$\mathbb{E}[\hat{\theta}^2] = \frac{T}{T - N - 2} \left(\theta^2 + \frac{N}{T} \right) = \frac{T\theta^2 + N}{T - N - 2}, \quad T > N + 2$$

- Sectors: $N = 9$, $T = 60$, true $\theta = 0.66$ a year: $\mathbb{E}[\hat{\theta}^2]$ corresponds to a Sharpe of 1.65 a year, the analytical counterpart of the simulated median 1.56

- Unbiased estimator of θ^2 : $\hat{\theta}_u^2 = [(T - N - 2)\hat{\theta}^2 - N] / T$

► It can be negative; truncating at zero gives a nonnegative but upward-biased estimator, and $\sqrt{\hat{\theta}_u^2}$ is not unbiased for θ

- Sectors, full sample ($T = 331$): plug-in $\hat{\theta} = 0.66$, bias-corrected $\sqrt{\hat{\theta}_u^2} = 0.31$ a year: more than half of the in-sample Sharpe is noise, even with 27 years

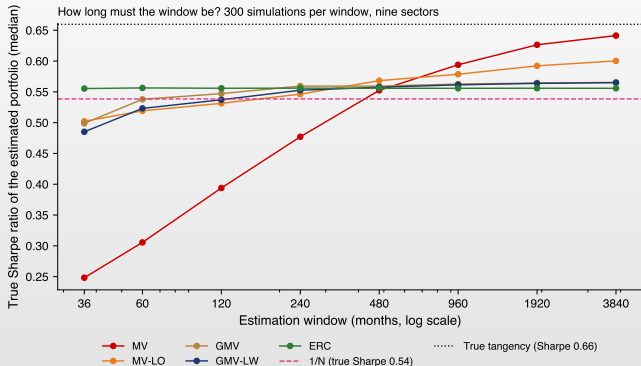
- The tangency weights are OLS coefficients of a regression of 1 on the returns (Britten-Jones (1999); [Appendix](#)), Seminar A2 and B9)

Unstable Weights



- Nine sector ETFs; each month the weights use only the previous 60 months; top panel clipped at ± 6
- MV: median gross exposure $\sum_i |w_i| = 4.9$, above 3 in 90% of the months; largest single weight 26 in absolute value
- Explosions in 2004–2006 and 2009: the denominator $1^\top \hat{\Sigma}^{-1} \hat{\mu}$ is close to zero, so the tangency weights have very heavy tails
 - Negative in 12 of 271 windows: MV then holds the minimum-Sharpe portfolio
- GMV-LW (bottom): gross exposure 1.5, largest weight 0.94
- Unstable weights mean high turnover, hence high trading costs

How Long Must the Window Be to Beat 1/N?



- DeMiguel, Garlappi & Uppal (2009): 14 models on 7 data sets; none beats 1/N consistently in Sharpe ratio, certainty-equivalent return or turnover; their calibration needs about 3000 months for 25 assets
- Here: 300 simulated samples per window, population parameters = sector moments
- MV beats 1/N only with 480 months (40 years): median true Sharpe 0.55 vs 0.54; with 60 months 0.31
- MV-LO, GMV, GMV-LW are close to 1/N from 60 months on; ERC is almost insensitive, 0.56
- In this calibration 1/N is nearly efficient (true tangency 0.66): simulation evidence, not an empirical test

 MFM_ch4_window_length

Why the Sample Covariance Matrix Fails

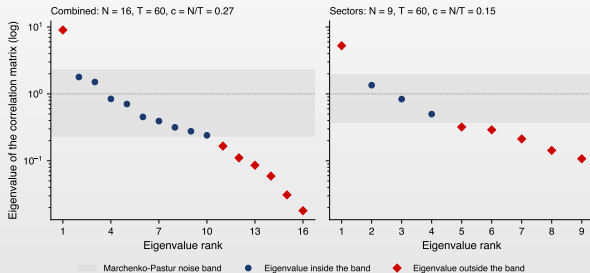
- ▣ Sample eigenvalues are too dispersed: the largest too large, the smallest too small
- ▣ **Condition number**: largest over smallest eigenvalue; a large value means Σ^{-1} amplifies noise
- ▣ Optimisers invert Σ , so they load on the directions with the smallest estimated variance
- ▣ The demeaned sample matrix has rank at most $T - 1$: singular, not invertible, whenever $T \leq N$
- ▣ **Marchenko–Pastur** benchmark ([Marčenko & Pastur \(1967\)](#)): i.i.d. returns with $\Sigma = \sigma^2 I$; as $N, T \rightarrow \infty$ with $N/T \rightarrow c \in (0, 1)$, the sample eigenvalues fill

$$[\lambda_-, \lambda_+] = [\sigma^2(1 - \sqrt{c})^2, \sigma^2(1 + \sqrt{c})^2]$$

- ▶ $c = 16/60 = 0.27$: band $[0.23, 2.30]$ for a correlation matrix ($\sigma^2 = 1$), a ten-fold spread from noise alone
- ▶ [Laloux et al. \(1999\)](#): eigenvalues outside the band are candidates for structure; those inside cannot be told from noise by this comparison alone

Sample Eigenvalues versus the Noise Band

Last 60-month window, Aug 2021 - Jul 2026



- Correlation matrices of monthly excess returns, last 60-month window (August 2021 – July 2026); shaded band: $[\lambda_-, \lambda_+]$ for $\sigma^2 = 1$
- Only the market mode lies above λ_+ : 9.0 (56% of the trace) for 16 ETFs, 5.2 for nine sectors ($\lambda_+ = 1.92$); with the noise variance rescaled to $\sigma^2 = 1 - \lambda_1/N$ (Laloux et al. (1999)), three eigenvalues exceed the band in each universe
- Six of 16 eigenvalues lie **below** the band: near-riskless combinations such as TLT against IEF; GMV loads exactly on them, so if they are underestimated, so is its risk

The Shrinkage Estimator

- Shrink the sample matrix S towards a structured target F

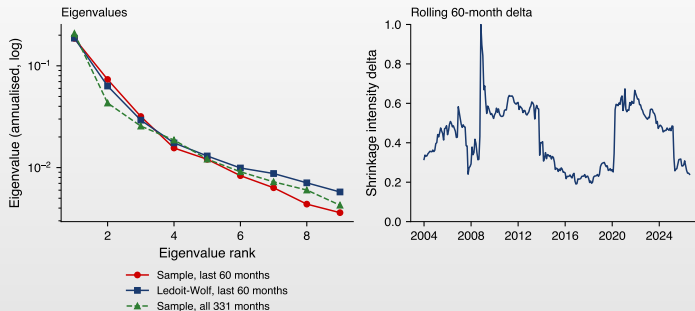
$$\hat{\Sigma}_{\text{LW}} = \delta F + (1 - \delta) S, \quad 0 \leq \delta \leq 1$$

- Targets

- ▶ Single-index (market) model: Ledoit & Wolf (2003)
- ▶ Scaled identity: Ledoit & Wolf (2004a)
- ▶ Constant correlation, $f_{ii} = s_{ii}$ and $f_{ij} = \bar{r} \sqrt{s_{ii} s_{jj}}$ for $i \neq j$, $\bar{r}$ the average sample correlation: Ledoit & Wolf (2004b), used here

- δ trades the variance of S against the bias of F : noisy data or a good target, large δ ; long samples, $\delta \rightarrow 0$ (the optimal δ^* in the [Appendix](#))

Shrinkage on US Sectors



- Nine sector ETFs, monthly excess returns; last window August 2021 – July 2026: $\delta = 0.24$; the smallest eigenvalues are pulled up, the largest almost unchanged; condition number $51.6 \rightarrow 32.5$
- Rolling δ : mean 0.43, range 0.19–1.00; $\delta = 1$ in the early windows, where 60 months are too few to beat the target
- GMV-LW: the GMV portfolio computed with $\hat{\Sigma}_{LW}$

Constraints Are Shrinkage Too

- ▣ Jagannathan & Ma (2003): the long-only GMV equals the unconstrained GMV of a modified covariance matrix

$$\tilde{S} = S - (\lambda \mathbf{1}^\top + \mathbf{1} \lambda^\top), \quad \mathcal{L} = \frac{1}{2} \mathbf{w}^\top S \mathbf{w} - \lambda_0 (\mathbf{w}^\top \mathbf{1} - 1) - \lambda^\top \mathbf{w}$$

- ▶ $\lambda \geq 0$: the Lagrange multipliers of the constraints $w_i \geq 0$ in $\mathcal{L}$; $\lambda_i > 0$ only where the constraint binds
- ▶ The first-order condition $S \mathbf{w} = \lambda_0 \mathbf{1} + \lambda$ and $\lambda^\top \mathbf{w} = 0$ give $\tilde{S} \mathbf{w} = \lambda_0 \mathbf{1}$: the GMV condition for $\tilde{S}$
- ▣ $\tilde{s}_{ij} = s_{ij} - \lambda_i - \lambda_j$: a binding constraint on asset i lowers its covariances, which shrinks the large estimates
- ▣ Imposing the “wrong” constraint (no short sales) can reduce out-of-sample risk; the effect is largest when covariances come from few observations

Nonlinear Shrinkage

- Linear shrinkage mixes two matrices with one δ ; **nonlinear** shrinkage keeps the sample eigenvectors and corrects each eigenvalue separately
 - ▶ Ledoit & Wolf (2017): tuned for portfolio selection; Ledoit & Wolf (2020): an analytical formula, no optimisation
- Out of sample, 60-month window, GMV volatility
 - ▶ Sectors: sample 11.90%, linear (constant correlation) 11.80%, nonlinear 11.62%
 - ▶ 16 ETFs: sample 4.06%, linear 4.38%, nonlinear 4.31%: better than the linear target, not than the raw matrix
- A constant correlation fits neither stock–stock (positive) nor stock–bond (negative) pairs: the target matters (Seminar 4, B4)
- Rule: shrink, constrain, or both, but never invert a raw sample matrix of many assets

From Covariance Shrinkage to Mean Shrinkage



Fischer Black (1938–1995)

Photo: Dalmatine at English

Wikipedia; public domain; Wikimedia

[Commons](#)

- Shrinkage fixes $\hat{\Sigma}$; Black & Litterman (1992), developed at Goldman Sachs, shrinks the means towards those implied by a **benchmark** portfolio w_b
- Reverse optimisation: a mean–variance investor with risk aversion $\delta_{\mathbf{RA}}$ and access to cash holds $w^* = (\delta_{\mathbf{RA}} \Sigma)^{-1} \mu$ in risky assets; the returns that make w_b optimal are

$$\pi = \delta_{\mathbf{RA}} \Sigma w_b$$

- π : **implied** (equilibrium) excess returns; with market weights, the CAPM returns
- Here: no market capitalisations for the sectors, so $w_b = 1/N$, $\delta_{\mathbf{RA}} = 2.5$
- Then add the investor's **views**, weighted by their confidence: the weights move away from the benchmark only where views exist

Views and the Posterior

▣ K views: $P\mu = q + \varepsilon$, $\varepsilon \sim N(0, \Omega)$; prior $\mu \sim N(\pi, \tau\Sigma)$

▣ Posterior mean: a precision-weighted average; weights $w_{BL} = (\delta_{RA}\Sigma)^{-1}\mu_{BL}$

$$\mu_{BL} = [(\tau\Sigma)^{-1} + P^\top \Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\pi + P^\top \Omega^{-1}q]$$

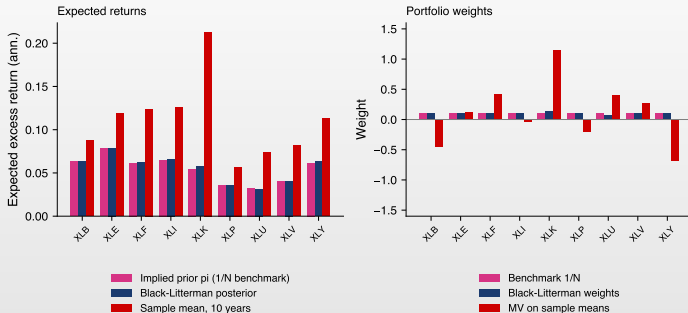
▣ Limits: $\Omega \rightarrow 0$ gives $P\mu_{BL} = q$ exactly; $\Omega \rightarrow \infty$ gives π ; without views, the benchmark itself

▣ He & Litterman (2002): $\tau = 0.05$, $\Omega = \text{diag}(P\tau\Sigma P^\top)$; with Σ in the weights, only the assets in the views move

▣ One asset, $\pi = 5\%$, $\tau\sigma^2 = 0.00128$, view 8% with variance 0.0004: $\mu_{BL} = 7.29\%$, view weight 0.76

▣ The same estimator is a GLS (mixed) regression (Theil & Goldberger (1961); [Appendix](#), Seminar A5)

Black–Litterman on US Sectors



- Σ from monthly excess returns, August 2016 – July 2026; one view: XLK minus XLU = 3% a year; implied 2.2%, posterior 2.6% (the average, since Ω equals the prior variance of the view)
- Weights: XLK from 11.1% to 14.4%, XLU from 11.1% to 7.8%, all others unchanged; the view is relative ($P1 = 0$), so the tilt $\propto P^\top$ sums to zero and the portfolio stays fully invested
- Sample means instead: XLK 21.3% a year; MV weights +115% XLK, -68% XLY, -46% XLB

Risk Contributions

- Volatility is homogeneous of degree one in the weights (Euler)

$$\sigma_p = \sum_{i=1}^N w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_{i=1}^N \frac{w_i (\Sigma w)_i}{\sigma_p}$$

- Risk contribution** of asset i : $RC_i = w_i (\Sigma w)_i / \sigma_p$; percentage share $RC_i / \sigma_p = w_i (\Sigma w)_i / \sigma_p^2$
- Worked example, 50/50 in SPY and IEF (monthly 2002–2026, annualised): $\sigma_1^2 = 0.02166$, $\sigma_2^2 = 0.00441$, $\sigma_{12} = -0.00098$
 - $(\Sigma w)_1 = 0.5(0.02166) + 0.5(-0.00098) = 0.01034$, $(\Sigma w)_2 = 0.5(-0.00098) + 0.5(0.00441) = 0.00172$
 - $\sigma_p^2 = 0.5(0.01034) + 0.5(0.00172) = 0.00603$, $\sigma_p = 7.76\%$; shares: SPY 85.8%, IEF 14.2%
- Equal weights are not equal risks: volatile assets dominate; risk budgeting chooses w so that the shares equal preset budgets b_i

Equal Risk Contribution

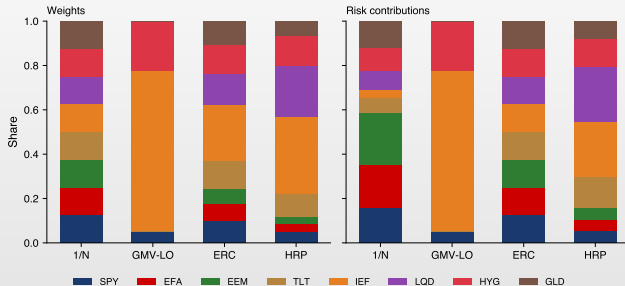
- ERC (Maillard, Roncalli & Teïletche (2010)): $RC_i = RC_j$ for all i, j ; computed from a convex problem, then normalised

$$\min_{y>0} \frac{1}{2} y^\top \Sigma y - \frac{1}{N} \sum_i \ln y_i, \quad w = y / 1^\top y$$

- For Σ positive definite a solution exists and is unique (strictly convex objective, infinite at the boundary)
- Volatility ordering: $\sigma_{\text{GMV}} \leq \sigma_{\text{ERC}} \leq \sigma_{1/N}$
- Two assets: ERC is **inverse volatility**, $w_1 = \sigma_1^{-1} / (\sigma_1^{-1} + \sigma_2^{-1})$, for any $\rho \in (-1, 1)$
 - SPY (14.7%) and IEF (6.6%): ERC holds 31% SPY, volatility 6.1%, each asset 50% of the variance
- Needs no expected returns, only Σ ; ERC is the tangency portfolio if all assets have the same Sharpe ratio and equal correlations

Weights versus Risk Contributions

Eight multi-asset ETFs, monthly May 2007 - Jul 2026

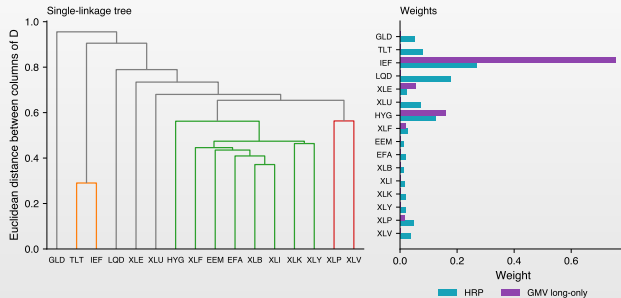


- Eight multi-asset ETFs, full-sample covariance matrix, May 2007 – July 2026 ($T = 231$)
- 1/N: equities and high yield receive half of the weight but 69% of the risk; ERC: 12.5% of the risk from every asset, IEF gets the largest weight (25%); long-only GMV: 73% IEF, 22% HYG, 5% SPY, five assets at zero
- Volatility: 1/N 9.4%, ERC 7.9%, HRP 7.1%, long-only GMV 5.8%: ERC lies between 1/N and GMV, as the theory predicts

Hierarchical Risk Parity: From a Tree to Weights

- HRP (López de Prado (2016)), three steps, no matrix inversion (works even when Σ is singular), all weights positive
 1. **Tree**: correlation distance $d_{ij} = \sqrt{(1 - \rho_{ij})/2} \in [0, 1]$; then the Euclidean distance between the **columns** of $D = (d_{ij})$, $\tilde{d}_{ij} = \sqrt{\sum_k (d_{ki} - d_{kj})^2}$; single linkage on $\tilde{d}_{ij}$ (repeatedly merge the two closest clusters) gives a **dendrogram**
 2. **Quasi-diagonalisation**: order the assets as the leaves of the tree, so that similar ones are adjacent
 3. **Recursive bisection**: split the **ordered list** in two halves; give each half weight in inverse proportion to its variance $\tilde{w}^\top \Sigma \tilde{w}$, with $\tilde{w}$ the inverse-variance weights inside the half; repeat inside each half
- Mantegna (1999): the same distance and the **minimum spanning tree** reveal the hierarchy of a stock market; textbook: Härdle & Simar (2019), cluster analysis
- A worked four-asset example: Seminar 4, B3

HRP on 16 ETFs



- Nine sectors and seven multi-asset ETFs, monthly, May 2007 – July 2026; closest pair TLT and IEF ($\tilde{d} = 0.29$), gold joins last (0.96)
- Equity cluster: cyclical sectors, EFA, EEM and high-yield credit; defensive sectors (utilities, staples, health care) and energy join it late
- HRP: all 16 weights positive, largest IEF 26.7%, LQD 17.6%, HYG 12.6%; long-only GMV: 75.6% in IEF and only five assets above zero

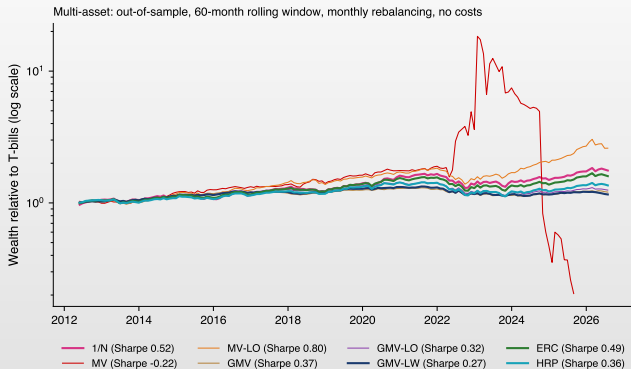
Backtest Design

- ▣ Rolling window of 60 months, monthly rebalancing; the weights of month t use months $t - 60, \dots, t - 1$ only
- ▣ Eight rules: 1/N, MV, MV-LO, GMV, GMV-LO, GMV-LW, ERC, HRP
- ▣ Three universes
 - ▶ Sectors: OOS January 2004 – July 2026, 271 months (first window January 1999 – December 2003)
 - ▶ Multi-asset and combined: OOS May 2012 – July 2026, 171 months
- ▣ Annualised mean, volatility and Sharpe ratio of excess returns; maximum drawdown of total-return wealth
- ▣ Every Sharpe ratio difference is reported with a test (next slides), and every test run enters the multiple-testing correction

Turnover and Costs: One Worked Month

- Between rebalancings weights drift with returns: $w_{i,t}^+ = w_{i,t}(1 + R_{i,t}) / \sum_j w_{j,t}(1 + R_{j,t})$
- **Turnover**: traded fraction of wealth, $TO_{t+1} = \sum_i |w_{i,t+1} - w_{i,t}^+|$
- Worked month, sectors, 1/N: in January 2004 XLB returned -4.4% and XLK $+4.3\%$, so the weights of 11.11% drifted to 10.50% and 11.45%
 - ▶ Rebalancing back to 1/N in February 2004 trades $TO = 0.019$: 1/N also trades, selling winners and buying losers
 - ▶ MV-LO, re-estimated on the new window, trades 0.748 in the same month
- Net return with cost c per unit traded: $R_{p,t}^{\text{net}} = R_{p,t} - c TO_t$; 1 basis point (bp) = 0.01%, so 20 bp means $c = 0.002$
 - ▶ At 20 bp the February trade costs MV-LO $0.002 \times 0.748 = 0.15\%$ of wealth, 1/N 0.004%

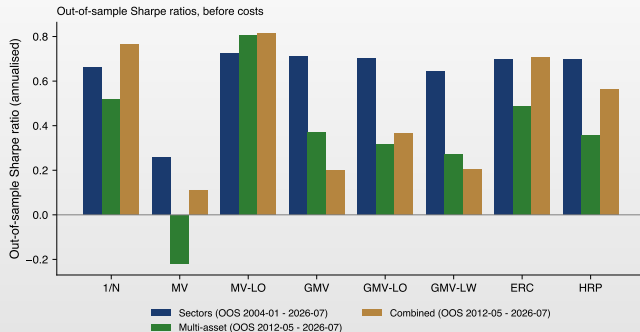
Out-of-Sample Growth: Multi-Asset



- Eight multi-asset ETFs; wealth relative to a Treasury-bill account, before costs
- MV collapses: volatility 518% a year, turnover 44 times wealth per month; a monthly loss above 100% ends its wealth path
- MV-LO has the highest Sharpe (0.80); 1/N 0.52; ERC 0.49
- GMV rules: lowest volatility (3.8%–5.4%), low Sharpe (0.27–0.37); they load on Treasuries, whose excess returns were slightly negative after May 2012 (TLT -0.5%, IEF -0.2% a year)
- Is MV-LO really better, or lucky? A test is needed

 MFM_ch4_oos_backtest

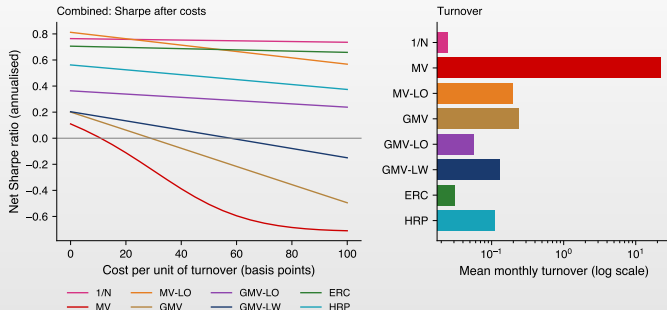
Sharpe Ratios Across Universes



- 60-month rolling window, monthly rebalancing, annualised, before costs; complete tables in the [Appendix](#)
- Sectors: every rule except MV lands within 0.07 of 1/N (0.66); MV 0.26 with drawdown -100%
- Combined: 1/N 0.76, MV-LO 0.81, ERC 0.71; the GMV rules 0.20–0.36
- MV: a monthly loss above 100% wipes out the capital; its mean and Sharpe describe a hypothetical return series, not investable wealth
- Turnover per month, sectors: 1/N 0.02, ERC 0.03, GMV-LW 0.14, MV-LO 0.23, MV 3.86

 [MFM_ch4_oos_backtest](#)

Sharpe Ratios after Costs



- Combined universe (16 ETFs); cost c from 0 to 100 bp per unit of turnover; turnover from 0.02 (1/N) to 22 (MV) per month
- Net Sharpe turns negative at about 11 bp for MV, 29 bp for GMV, 58 bp for GMV-LW; MV-LO overtakes 1/N only for costs below about 22 bp (Seminar 4, B8)
- 1/N and ERC barely move: low turnover is a hidden advantage; report net performance for realistic costs, not only gross

Inference on One Sharpe Ratio: Lo (2002)

- Lo (2002), i.i.d. Normal returns: $\sqrt{T}(\widehat{SR} - SR) \rightarrow N(0, 1 + SR^2/2)$, with SR monthly
- 1/N on sectors, $T = 271$: monthly $\widehat{SR} = 0.191$, standard error 0.061
 - ▶ Annualised $0.66 \pm 1.96 \times 0.21$: 95% interval $[0.25, 1.08]$
- Annualising by $\sqrt{q}$ ($q = 12$) is exact only without autocorrelation; in general $SR(q) = \eta(q) SR$

$$\eta(q) = \frac{q}{\sqrt{q + 2 \sum_{k=1}^{q-1} (q-k) \rho_k}}$$

- Here $\rho_1 = -0.03$, $\sum_{k=1}^{11} \rho_k = -0.11$: $\eta = 3.62$ instead of $\sqrt{12} = 3.46$, annual Sharpe 0.69

Testing a Difference in Sharpe Ratios

- H_0 : $SR_1 = SR_2$ for two strategies evaluated on the same months; [Jobson & Korkie \(1981\)](#): a test under i.i.d. Normal returns
- [Ledoit & Wolf \(2008\)](#): a robust version
 - ▶ $\hat{\Delta} = f(\hat{v})$, $v = (\mu_1, \mu_2, \mathbb{E}[r_1^2], \mathbb{E}[r_2^2])$, $f = \frac{\mu_1}{\sqrt{\mathbb{E}r_1^2 - \mu_1^2}} - \frac{\mu_2}{\sqrt{\mathbb{E}r_2^2 - \mu_2^2}}$
 - ▶ Standard error by the delta method: $\sqrt{\nabla f^\top \hat{\Psi} \nabla f / T}$, $\hat{\Psi}$ a HAC covariance
- Strategies on the same assets are highly correlated: small differences can be significant

The Block Bootstrap

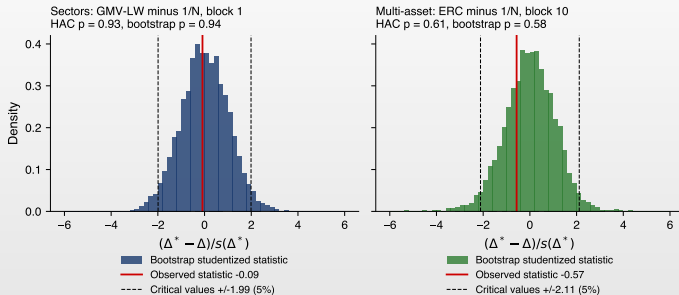


Bradley Efron, inventor of the bootstrap (1979)

Photo: Ryan K. Morris (2007); public domain; Wikimedia Commons

- Bootstrap: resample the data to approximate the sampling distribution of $\hat{\Delta}$
- Returns are dependent (volatility clustering): resample **blocks** of consecutive months (Künsch (1989))
- Ledoit & Wolf (2008), Section 3.2.2: circular blocks of **pairs** (r_{1t}, r_{2t}) , wrapping around the end of the sample, so the correlation of the two strategies is kept
- Centre and studentize: bootstrap $|\hat{\Delta}^* - \hat{\Delta}|/s(\hat{\Delta}^*)$; asymptotically pivotal, more accurate than the Normal approximation of the HAC test
- $p = [\#\{\tilde{d}^{*m} \geq |\hat{\Delta}|/s(\hat{\Delta})\} + 1]/(M + 1)$, $M = 4999$ draws
- Block length chosen by the calibration of Ledoit & Wolf (2008), Algorithm 3.1 (Appendix): mostly 1 for sectors, 8–10 for multi-asset and combined rules

Is Anything Better than $1/N$?



- Bootstrap distributions of the studentized difference against $1/N$; red: observed statistic; dashed: bootstrap critical values
- Sectors, GMV-LW minus $1/N$: -0.02 , 95% interval $[-0.36, 0.33]$, HAC $p = 0.93$, bootstrap $p = 0.94$ (block 1); multi-asset, ERC minus $1/N$: -0.03 , $[-0.15, 0.08]$, HAC $p = 0.61$, bootstrap $p = 0.58$ (block 10): so correlated with $1/N$ that the interval is narrow, yet it covers zero
- Sectors: no rule differs from $1/N$ at 5% (closest ERC, $+0.04$, HAC $p = 0.09$); multi-asset: MV-LO $+0.29$ is **not** significant (HAC $p = 0.12$, bootstrap 0.16)

Many Rules, Many Tests

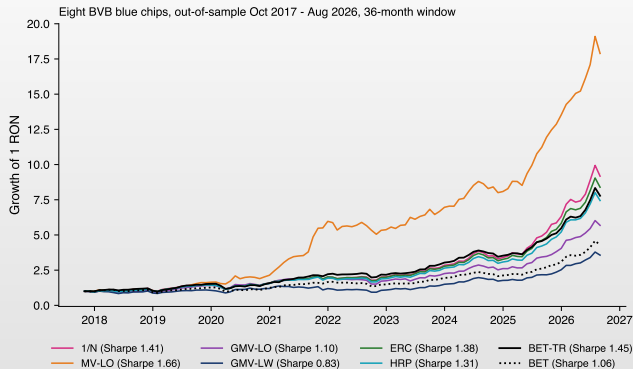
Universe	Rule minus 1/N	$\hat{\Delta}$	HAC p	Bootstrap p
Multi-asset	MV	-0.74	0.01	0.06
Multi-asset	HRP	-0.16	0.07	0.05
Combined	GMV	-0.57	0.03	0.05
Combined	GMV-LO	-0.40	0.05	0.08
Combined	GMV-LW	-0.56	0.02	0.03
Combined	MV-LO	+0.05	0.82	0.82

- No rule beats 1/N significantly in any universe; unadjusted, GMV rules lose to 1/N on the combined universe: low risk, too little return
- The family: $H_0: \Delta_k = SR_k - SR_{1/N} \leq 0$ for all k ; 21 tests (7 rules $\times$ 3 universes); Holm-adjusted p -values are at least 0.29 (HAC) and 0.68 (bootstrap): nothing survives ([Chapter 3](#), multiple testing)
- A different question, the best of 24 configurations: the deflated Sharpe ratio of [Bailey & López de Prado \(2014\)](#) gives $0.84 < 0.95$ for MV-LO on 16 ETFs, but the verdict hinges on the effective number of trials ([Appendix](#))

Setup: a Romanian Investor

- ▣ Eight blue chips listed since 2014: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL
- ▣ Monthly returns in RON, October 2014 – August 2026; 36-month window, first portfolio October 2017
- ▣ Rules: 1/N, MV-LO, GMV-LO, GMV-LW, ERC, HRP; benchmarks BET-TR and BET
- ▣ Sharpe ratios with $R_f = 0$ (no RON money-market series in the data): return per unit of risk, higher than excess-return Sharpe ratios
- ▣ Earlier Romanian evidence on diversification: [Dragotă, Șerbănescu & Pele \(2008\)](#)

BVB Portfolios versus BET-TR



- Out of sample, October 2017 – August 2026 (107 months), before costs, $R_f = 0$
- BET-TR: mean 24.7% a year, volatility 17.0%, Sharpe 1.45; BET: mean 17.8%, Sharpe 1.07; the gap of about 7% a year is the dividends: compare with BET-TR, never with BET
- 1/N 1.41, ERC 1.38, HRP 1.31: close to BET-TR
- MV-LO 1.66 (difference $+0.21$, $p = 0.44$); GMV-LW 0.83 (-0.63 , $p = 0.05$)
- Average correlation of the eight stocks 0.44: little room for diversification inside the BVB; the latest GMV-LO puts 70% in TLV
- Open question: does diversification **beyond** BET pay? (Seminar 4, C1)

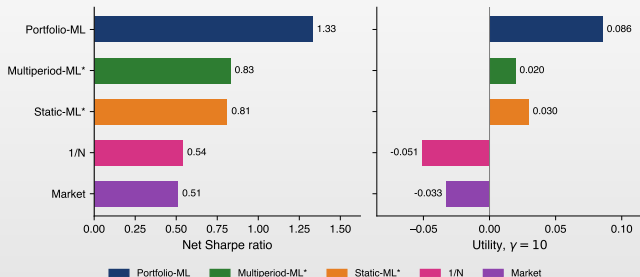
Case Study: Which Frontier Can a Large Investor Reach?

- Jensen, Kelly, Malamud & Pedersen (2026), *Review of Financial Studies*: how should portfolios be built from machine-learning return forecasts when every trade moves prices?
- **Implementable efficient frontier** (eq. 12): for each volatility σ , the highest expected return **net of trading costs**, $k(\sigma) = \max_{\pi} E[r_{t+1}^{\text{net}}]$ s.t. $E[\pi_t^\top \Sigma \pi_t] = \sigma^2$
 - Proposition 1: the net Sharpe ratio $k(\sigma)/\sigma$ falls along the frontier, and a larger investor faces a lower frontier
- Cost as a fraction of wealth W_t (eqs. 4, 7): $c_t = \frac{W_t}{2}(\pi_t - g_t \pi_{t-1})^\top \Lambda (\pi_t - g_t \pi_{t-1})$, quadratic in the trade; g_t drifts last month's weights, $\Lambda = \text{diag}(0.2/V_i)$, V_i the dollar volume
- Optimal rule: trade only part of the way towards an **aim portfolio**, $\pi_t = m g_t \pi_{t-1} + (1 - m)A_t$ (Gârleanu & Pedersen (2013)); Portfolio-ML learns A_t from the signals by a ridge regression that maximises utility net of costs

	Paper	Course replication
Assets	NYSE stocks above median size (184–805)	14 US stocks + 9 sector ETFs
Signals	115 characteristics (Jensen, Kelly & Pedersen (2023))	5 price and volume signals
Out of sample	1981–2020, monthly	February 2014 – August 2026 (151 months)
Investor	$\gamma = 10$, wealth \$10 billion at the end of 2020	same

Case Study: Table 2 at a Glance

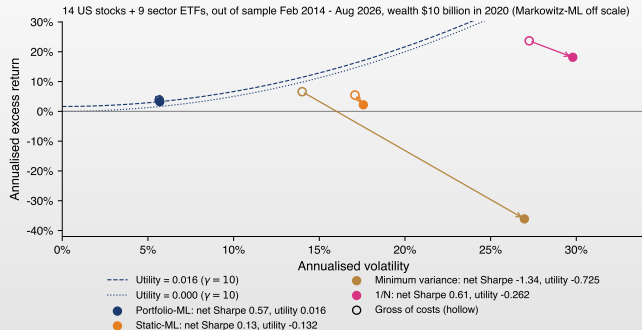
Out of sample 1981–2020, US stocks, wealth \$10 billion (Markowitz-ML off scale)



- Jensen, Kelly, Malamud & Pedersen (2026), Table 2: net of costs, 1981–2020, $\gamma = 10$; utility (eq. 11) = annual net return $-\frac{\gamma}{2} \times$ annual variance
- Markowitz-ML (mean–variance on ML return forecasts, no costs): gross Sharpe 2.06, turnover 79.36, leverage 74.61: net return extremely negative
- Portfolio-ML: net Sharpe 1.33, utility 0.086 (net return 15%, volatility 11%, cost 0.6% a year)
- Static-ML* (one-period costs; * = extra tuning layer) 0.81, 0.030; 1/N 0.54, -0.051 ; market 0.51, -0.033 : only cost-aware rules have positive utility

 MFM_ch4_implementable_frontier

Case Study: Gross and Net of Costs on the Course Data



- Hollow marker: before costs; filled: after the costs of eq. (35); dashed curve: pairs with the utility of Portfolio-ML (0.016), dotted: utility 0
- Portfolio-ML barely moves: 3.8% gross, 3.3% net, 5.7% volatility; net Sharpe 0.57
- 1/N keeps 18.1% net (Sharpe 0.61), but at 29.8% volatility its utility is -0.26
- Minimum variance: +6.6% gross, -36.1% net, rebalancing a leverage of 2.6 costs 43% a year
- Limits: 23 assets, 5 signals, no factor model for Σ ; the frontier that matters is the one after costs (Seminar 4, C3)

 MFM_ch4_implementable_frontier

An Open Question

- Does any **estimated** allocation rule beat $1/N$ out of sample, net of costs, once every rule, window and universe tried is counted?
- Live evidence points both ways
 - ▶ Nonlinear shrinkage of Σ improves GMV out of sample in large cross-sections: [Ledoit & Wolf \(2017\)](#)
 - ▶ Combining tangency, GMV and $1/N$ under estimation risk: [Kan, Wang & Zhou \(2022\)](#)
 - ▶ Backtests need haircuts for the number of specifications tried: [Harvey & Liu \(2015\)](#)
- Formal statement for K rules on one universe: $H_0: \Delta_k = SR_k - SR_{1/N} \leq 0$ for all k ; $H_1: \Delta_k > 0$ for some k ; error rate fixed for the whole family
- One pass of the loop with an AI assistant: literature with DOIs, competing hypotheses, code on the course's monthly data, a pre-registered family of tests, a referee-style critique (prompts in the [Appendix](#))
- Falsified by: a positive $\hat{\Delta}_k$ that survives a family-wise correction, costs and a second sample

What the Researcher Must Check

- **Look-ahead:** weights at t built with month- t returns; assert that the last estimation date is before t
- **Survivorship and selection:** an ETF list chosen today; use the universe available at each date, including later delisted ETFs, or disclose the limitation
- **Multiple testing:** 7 rules $\times$ 3 universes already give 21 tests; Holm (FWER) or Benjamini–Hochberg (FDR, [Benjamini & Hochberg \(1995\)](#)) adjusted p -values
- **Power:** $\hat{\Delta}$ has a standard error near 0.18 on 171 months; compute the detectable effect before running
- **Return type:** portfolios aggregate **simple** returns, Sharpe ratios use **excess** returns; recompute one month step by step
- **Invented references:** an unresolvable DOI or a title mismatch drops the claim and goes into `AI_ERRORS.md`

Mini-Case: Data Snooping in This Chapter's Backtest

- ▣ A typical assistant proposal: “MV-LO beats $1/N$ on the multi-asset universe”; superiority is one-sided: $H_0: \Delta \leq 0$, $p = 1 - \Phi(\hat{\Delta}/\text{se})$
- ▣ Replicate: $\hat{\Delta} = +0.29$, HAC standard error 0.18, one-sided $p = 0.06$ (two-sided HAC 0.12, bootstrap 0.16)
- ▣ Test the family, not the winner: 21 tests (7 rules $\times$ 3 universes)
 - ▶ Only 7 of 21 differences are positive; the 4 two-sided HAC p -values below 0.05 are all **losses**, which never reject a superiority null
 - ▶ One-sided: smallest $p = 0.046$ (sectors, ERC); adjusted Holm 0.96, Benjamini–Hochberg 0.62; 0 discoveries at an FDR of 10%
- ▣ Power 0.8 for $\Delta = 0.29$, one-sided: about 435 months (36 years) at 5%, 944 months (79 years) at the Holm level 5%/21
- ▣ Interpretation: the data cannot rank these rules; the bootstrap interval for MV-LO, $[-0.13, 0.70]$, admits losses and large gains; failure to reject is not proof of equality

Your Turn: A Research Project Idea

- ▣ **Question:** does nonlinear shrinkage (Ledoit & Wolf (2017)) or a three-fund combination (Kan, Wang & Zhou (2022)) beat $1/N$ on the 16 ETFs of this chapter, after 10–50 bp costs?
- ▣ **Replicate first:** the combined-universe results ($1/N$ Sharpe 0.76, MV-LO 0.81) and the adjusted p -values of the mini-case
- ▣ **Pre-register:** hypotheses, family of tests, error rate, power and a placebo (random long-only weights with the same turnover), fixed before any result
- ▣ **Report:** code on the course's monthly data, a table of all specifications, a referee-style self-critique
- ▣ **Log:** prompts in AI_USE.md; every error caught and the test that caught it in AI_ERRORS.md

Conclusions: Which Rule, When?

Rule	Objective	Inputs	Main failure mode	Evidence in this chapter
MV	maximum Sharpe	$\hat{\mu}, \hat{\Sigma}^{-1}$	error maximisation, explosive weights	worst in every universe; insolvent path
MV-LO	maximum Sharpe, $w_i \geq 0$	$\hat{\mu}, \hat{\Sigma}$	corner solutions, turnover	best point estimates, never significant
GMV, GMV-LW	minimum variance	$\hat{\Sigma}$ (shrunk)	loads on low-return assets	lowest risk, loses to 1/N on 16 ETFs (unadjusted)
ERC, HRP	balanced risk	$\hat{\Sigma}$	a tilt of 1/N; HRP trades more	never significantly better than 1/N
Black–Litterman	benchmark plus views	w_b , views, Ω	confidence set too high	moves only the assets in the views
1/N	none	none	ignores risk differences	hard to beat after costs

- Evaluate every rule out of sample, net of costs, with a test of the Sharpe difference and a count of all tests; these verdicts are specific to these samples

Check Yourself, and Next

Retrieval questions

- Why does the in-sample tangency Sharpe overstate the achievable one, and by how much for $N = 9$, $T = 60$?
- Why does a long-only constraint act like shrinkage?
- Why can a small Sharpe difference between ERC and $1/N$ be nearly significant while a larger one between GMV and $1/N$ is not?

Links to other chapters

- **Chapter 3**: the tangency portfolio and the CAPM; factor covariance models
- **Chapters 5–6**: time-varying volatility and correlation for dynamic weights
- **Chapter 7**: portfolio Value-at-Risk and Expected Shortfall
- **Chapter 8**: backtesting without overfitting
- **Chapter 13**: machine learning for returns and portfolios

Appendix: Wishart Moments and the Distribution of Weights

- i.i.d. $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ returns, maximum-likelihood estimators: $\hat{\boldsymbol{\mu}} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}/T)$, $T\hat{\boldsymbol{\Sigma}} \sim W_N(T-1, \boldsymbol{\Sigma})$ (Wishart), independent
- Inverse Wishart moment, finite for $T > N + 2$: $\mathbb{E}[\hat{\boldsymbol{\Sigma}}^{-1}] = \frac{T}{T-N-2}\boldsymbol{\Sigma}^{-1}$, an **upward** bias that grows with N/T
- Conditioning on $\hat{\boldsymbol{\mu}}$ and using $\mathbb{E}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] = \text{tr}(\mathbf{A} \text{Var}(\mathbf{x})) + \mathbb{E}[\mathbf{x}]^\top \mathbf{A} \mathbb{E}[\mathbf{x}]$ gives $\mathbb{E}[\hat{\theta}^2]$ of the lecture (Seminar A9)
- GMV weights with Wishart $\hat{\boldsymbol{\Sigma}}$ and T large enough relative to N : $\hat{\mathbf{w}}_{\text{GMV}}$ is unbiased and its exact distribution and covariance are known (Okhrin & Schmid (2006))
- Tangency weights divide by $\mathbf{1}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}$, which can be near zero: very heavy tails
- Estimation-risk-optimal mixtures of tangency, GMV and $1/N$: Kan & Zhou (2007), Kan, Wang & Zhou (2022)

Appendix: Tangency Weights by Regression, Britten-Jones (1999)

- Britten-Jones (1999): regress the constant 1 on the excess returns, **without** a constant term

$$1 = \mathbf{b}^\top \mathbf{r}_t + u_t, \quad \hat{\mathbf{b}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{1}_T \propto \hat{\Sigma}^{-1} \hat{\boldsymbol{\mu}}$$

- Sherman–Morrison on $\mathbf{X}^\top \mathbf{X} / T = \hat{\Sigma} + \hat{\boldsymbol{\mu}} \hat{\boldsymbol{\mu}}^\top$ gives the proportionality (Seminar A2); $\hat{\mathbf{b}} / \mathbf{1}^\top \hat{\mathbf{b}} =$ sample tangency weights if $\mathbf{1}^\top \hat{\mathbf{b}} > 0$
- OLS t -test of $b_i = 0$: asset i has zero weight in the tangency portfolio; F -test of $\mathbf{R} \mathbf{b} = 0$ with $\mathbf{R} = [\mathbf{I}_{N-1}, -\mathbf{1}_{N-1}]$: “the tangency portfolio is $1/N$ ”
- Exact finite-sample t and F distributions under i.i.d. Normal returns; HAC standard errors otherwise
- Nine sectors, January 1999 – July 2026 ($T = 331$): normalised $\hat{\mathbf{b}}$ reproduces the tangency weights; largest $|t| = 1.12$ (financials), HAC 1.19
- H_0 : tangency = $1/N$, eight restrictions: $F(8, 322) = 0.48$, $p = 0.87$; HAC Wald $\chi_8^2 = 4.50$, $p = 0.81$; not rejecting is not proof of equal weights

Appendix: The Optimal Shrinkage Intensity

- Ledoit–Wolf choose δ to minimise the expected squared (Frobenius) distance to the true Σ

$$\delta^* = \max \left\{ 0, \min \left\{ 1, \frac{\hat{\kappa}}{T} \right\} \right\}, \quad \hat{\kappa} = \frac{\hat{\pi} - \hat{\rho}}{\hat{\gamma}}$$

- $\hat{\pi}$: sum of the asymptotic variances of the entries of S (noise)
- $\hat{\rho}$: their asymptotic covariances with the target entries
- $\hat{\gamma} = \|F - S\|^2$: misspecification of the target
- Complete formulas for the constant-correlation target: [Ledoit & Wolf \(2004b\)](#), Appendix; code: the [shrinkage Quantlet](#)

Appendix: Shrinking the Means and Other Covariance Estimators

- ▣ Bayes–Stein ([Jorion \(1986\)](#)): $\hat{\mu}_{BS} = (1 - \hat{\phi}) \hat{\mu} + \hat{\phi} \hat{\mu}_{GMV}$
 - ▶ Empirical-Bayes shrinkage of every mean towards the mean of the GMV portfolio; $\hat{\phi}$ from a Stein argument, larger when N/T is large
- ▣ [Tu & Zhou \(2011\)](#): the convex combination of a sophisticated rule and $1/N$ that minimises the expected out-of-sample loss
- ▣ [DeMiguel, Garlappi, Nogales & Uppal \(2009\)](#): norm constraints $\|w\|_1 \leq \delta$ or $\|w\|_2 \leq \delta$ nest the long-only GMV, shrinkage GMV and $1/N$; the bridge to the lasso portfolios of [Chapter 13](#)
- ▣ Factor structure plus a sparse residual covariance: POET ([Fan, Liao & Mincheva \(2013\)](#)), the link to [Chapter 3](#)
- ▣ High-dimensional GMV with shrinkage towards a target portfolio: [Bodnar, Parolya & Schmid \(2018\)](#)

Appendix: Black–Litterman as Mixed (GLS) Estimation

- ▣ Theil & Goldberger (1961): stack the prior and the views as one regression on μ

$$\begin{pmatrix} \pi \\ q \end{pmatrix} = \begin{pmatrix} I \\ P \end{pmatrix} \mu + \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}, \quad \text{Var} \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} = \begin{pmatrix} \tau \Sigma & 0 \\ 0 & \Omega \end{pmatrix}$$

- ▣ GLS gives exactly μ_{BL} , with covariance $M = [(\tau \Sigma)^{-1} + P^\top \Omega^{-1} P]^{-1}$
- ▣ Equivalent form: $\mu_{BL} = \pi + \tau \Sigma P^\top (P \tau \Sigma P^\top + \Omega)^{-1} (q - P \pi)$
- ▣ $\Omega \rightarrow 0$: restricted least squares, $P \mu_{BL} = q$
- ▣ Predictive covariance $\Sigma + M$ in the weights would change all weights; without views it scales them by $1/(1 + \tau)$

Appendix: Other Risk-Based Rules

- ▣ **Inverse volatility:** $w_i \propto 1/\sigma_i$; equals ERC when all correlations are equal
- ▣ **Maximum diversification** (Choueifaty & Coignard (2008)): maximise $w^\top \sigma / \sqrt{w^\top \Sigma w}$, the ratio of the average asset volatility to the portfolio volatility
- ▣ **Minimum variance:** GMV, the extreme risk-based rule
- ▣ All ignore expected returns: robust to estimation error, but optimal only under special assumptions

Appendix: Results, Sector Universe

Rule	Mean	Volatility	Sharpe	Turnover	Max. drawdown
1/N	9.4%	14.3%	0.66	0.02	−49%
MV	17.8%	69.0%	0.26	3.86	−100%
MV-LO	10.7%	14.7%	0.72	0.23	−42%
GMV	8.5%	11.9%	0.71	0.23	−33%
GMV-LO	8.1%	11.5%	0.70	0.09	−34%
GMV-LW	7.6%	11.8%	0.65	0.14	−31%
ERC	9.3%	13.2%	0.70	0.03	−47%
HRP	8.9%	12.8%	0.70	0.08	−45%

- January 2004 – July 2026; turnover: mean monthly traded fraction of wealth; drawdown of total-return wealth
- MV: a monthly loss above 100% wipes out the capital (drawdown −100%); its mean and Sharpe describe a hypothetical return series

Appendix: Results, Combined Universe (16 ETFs)

Rule	Mean	Volatility	Sharpe	Turnover	Sharpe, 50 bp
1/N	7.9%	10.3%	0.76	0.02	0.75
MV	28.7%	261.8%	0.11	22.10	-0.51
MV-LO	8.0%	9.8%	0.81	0.20	0.69
GMV	0.8%	4.1%	0.20	0.24	-0.15
GMV-LO	2.0%	5.4%	0.36	0.06	0.30
GMV-LW	0.9%	4.4%	0.20	0.13	0.03
ERC	5.5%	7.8%	0.71	0.03	0.68
HRP	4.0%	7.0%	0.56	0.11	0.47

▣ May 2012 – July 2026; last column: net of 50 basis points per unit of turnover

Appendix: Bootstrap Details and the Block Length

- Studentization: $s(\hat{\Delta}^*)$ from the block sums of each resample; $s(\hat{\Delta})$ on the data from a prewhitened quadratic-spectral kernel ([Andrews & Monahan \(1992\)](#)) with automatic bandwidth ([Andrews \(1991\)](#))
- [Ledoit & Wolf \(2008\)](#), Algorithm 3.1: choose the block b whose bootstrap interval has actual coverage closest to 95%
 - ▶ Fit a first-order vector autoregression to the pair of returns; resample its residuals by a stationary bootstrap ([Politis & Romano \(1994\)](#)), mean block 5
 - ▶ Generate $K = 5000$ pseudo-samples of length T ; in each, build the 95% interval for $b \in \{1, 2, 4, 6, 8, 10\}$
 - ▶ $\hat{g}(b)$: share of intervals that contain the pseudo-true $\hat{\Delta}$; pick $\tilde{b} = \arg \min_b |\hat{g}(b) - 0.95|$
- The grid and K are those of the paper (Sections 4–5); applied to each of the 21 pairs of this chapter
- A non-studentized bootstrap with a fixed 6-month block is not this test: its block is not calibrated for coverage

Appendix: The Deflated Sharpe Ratio

- Bailey & López de Prado (2014): the best of N Sharpe ratios is biased upwards even if every true Sharpe ratio is zero
- Expected maximum under H_0 : all true Sharpe ratios zero, N independent trials, monthly units as T ; with $\gamma_E = 0.5772$ (Euler's constant) and $\hat{V}$ the variance of the N estimates

$$SR_0 = \sqrt{\hat{V}} \left[(1 - \gamma_E) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma_E \Phi^{-1} \left(1 - \frac{1}{Ne} \right) \right]$$

- Deflated Sharpe ratio, with skewness $\hat{\gamma}_3$ and kurtosis $\hat{\gamma}_4$ of the winner

$$DSR = \Phi \left(\frac{(\widehat{SR} - SR_0) \sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right)$$

- This chapter: 24 configurations (8 rules $\times$ 3 universes); winner MV-LO on 16 ETFs, 0.81 a year
 - As 24 independent trials: $SR_0 = 0.52$ a year, $DSR = 0.84 < 0.95$
 - The configurations are strongly correlated (up to 0.99), so fewer are independent: $N = 8$ gives DSR 0.93, $N = 3$ gives 0.98
- Holm adjusts the whole family of pairwise tests against $1/N$; the DSR asks whether the single best configuration is better than luck

Appendix: Beyond Equities, Digital Assets

- ▣ [Petukhina, Trimborn, Härdle & Elendner \(2021\)](#): cryptocurrencies in portfolio allocation rules; gains depend on the rule and on liquidity
- ▣ [Pele et al. \(2023\)](#): are cryptos becoming alternative assets?
- ▣ Bitcoin: annualised volatility 172% (monthly, 2010–2026, one surviving coin chosen ex post), IEF 6.6%
- ▣ Risk contributions grow with volatility: risk-based rules give Bitcoin a small weight automatically (Seminar 4, B5)
- ▣ Robust mean–variance for crypto-assets: [Pele \(2023\)](#)

Appendix: Case Study, Method in Detail (1/2)

- Cost (eqs. 4, 7): π_t risky-asset weights, $1 - \mathbf{1}^\top \pi_t$ in the risk-free asset; $g_t = \text{diag}((1 + r_t^f + r_t)/(1 + g_t^w))$ (eq. 6), g_t^w the growth of wealth; $\Lambda = \text{diag}(0.2/V_i)$, V_i the average daily dollar volume over 6 months: trading 1% of V_i moves the price by 0.1% (eq. 35, [Frazzini, Israel & Moskowitz \(2018\)](#))
- Optimal rule (Proposition 3): $\pi_t = m g_t \pi_{t-1} + (1 - m) A_t$, A_t a weighted sum of today's and expected future Markowitz portfolios; m solves eq. (14); [Gârleanu & Pedersen \(2013\)](#) generalised
- Portfolio-ML (Proposition 4, eq. 40): $A_t = \text{diag}(1/\sigma_{i,t}) f(s_t) \beta$, $f = p$ random Fourier features of the signals s_t ; $\hat{\beta} = (E_T[\tilde{\Sigma}_t] + \lambda I)^{-1} E_T[\tilde{r}_{t+1}]$, a ridge regression that maximises utility net of costs in closed form (eqs. 24–26), learning the weights directly as [Brandt, Santa-Clara & Valkanov \(2009\)](#) do without costs

Appendix: Case Study, Method in Detail (2/2)

- Data (Section 5.1): NYSE common stocks above the median market capitalisation (184–805 stocks), 115 characteristics of Jensen, Kelly & Pedersen (2023) ranked into $[0, 1]$; December 1952 – December 2020, out of sample 1981–2020, monthly
- Σ : 13 characteristic themes and 12 industries as factors, exponential weights with half-lives of 378 days (correlations) and 126 days (variances); investor $\gamma = 10$, wealth \$10 billion at the end of 2020, growing with the market (Section 5.1.3)
- Tuning (Table 1): $\lambda \in \{0, e^{-10}, e^{-9.8}, \dots, e^{10}\}$, $p \in \{2^6, \dots, 2^9\}$, $\eta \in \{e^{-3}, e^{-2}\}$ (standard deviation of the random weights), chosen each year by the utility of validation backtests since 1971
- Rivals: Markowitz-ML (eq. 32), Static-ML (one-period cost, eq. 34), Multiperiod-ML (forecasts for 12 horizons, eq. 22), factor sorts, 1/N, minimum variance, market; “*” marks an extra tuning layer (Section 5.3)

Appendix: Case Study, Complete Results

- Table 2, Portfolio-ML: net Sharpe ratio 1.33, utility 0.086 (net return 15%, volatility 11%, cost 0.6% a year, turnover 0.25)
- Table 2, Multiperiod-ML*: net Sharpe 0.83, utility 0.020; Portfolio-ML needs no second tuning layer
- Table 3: posterior probability that Portfolio-ML has the higher utility: 99.9% against Static-ML* and Multiperiod-ML*, 100% against Markowitz-ML
- Figure 1(B) (Section 6.2): with \$10 billion and $\gamma = 10$, net excess return 14.6% at 11% volatility; with \$1 billion, 16.3% at the same volatility: size is a cost
- Figure 5: net of costs, quality is the most important theme and short-term reversal almost irrelevant; monthly autocorrelation -0.04 for 1-month reversal, 0.94 for book-to-market (Section 6.3): persistent signals are cheap to trade
- Limits stated by the authors: fees and other frictions, computing power and ML methods not available in real time early in the sample (Section 6.1); costs and risk treated as constant over the horizon of each optimisation (Section 5.1.3); among the smallest 20% of stocks Portfolio-ML ranks behind Multiperiod-ML* and Static-ML* (Figure 8)

Appendix: Case Study, the Course Replication

- Course data: 14 US stocks and 9 sector ETFs; five signals (1-month return, momentum from $t - 6$ and from $t - 12$ to $t - 1$, 1-year volatility, 6-month dollar volume); same cost rule, γ , wealth and grid; Σ with the same weights on the 23 assets; validation from 2011; out-of-sample returns February 2014 – August 2026 (151 months)
- Portfolio-ML: net Sharpe 0.57, utility 0.016, cost 0.6% a year, leverage 0.32; 1/N: net Sharpe 0.61, utility -0.26 (net volatility 30%)
- Utility: Static-ML -0.13 ; minimum variance -0.72 (cost 43% a year); Markowitz-ML extremely negative (leverage 18)
- Posterior probability of a higher utility for Portfolio-ML: 99.5% against Static-ML, 99.9% against minimum variance, 93.3% against 1/N
- Same ranking of cost-aware against cost-agnostic rules as Table 2; Portfolio-ML wins with small, slowly traded positions, not with a higher gross Sharpe ratio (0.68 against 0.87 for 1/N)
- Limits: 23 assets instead of 184–805, five signals instead of 115, no factor model for Σ , whose window reaches 10 years only in August 2014 (common prices from August 2004); return forecasts with $p \geq 2^6$ and three validation years (paper: 2^1 – 2^9 , ten years), no starred rivals
- Open extension: implementable frontiers for wealth of \$1, 10 and 100 billion and $\gamma \in \{1, 5, 10, 20, 100\}$, with 1/N's trading costs computed at each wealth level (Seminar 4, C3)

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List peer-reviewed studies since 2015 comparing shrinkage or risk-based rules with $1/N$ out of sample; for each give DOI, N , T , cost assumption and test."* Resolve every DOI on Crossref and open the table cited
- **Hypotheses:** *"Give three competing explanations for a Sharpe gain over $1/N$ (better Σ , a low-volatility premium, luck) and the testable implication of each for turnover, subperiods and factor loadings."*
- **Data and code:** *"Write Python that reads the course's daily price files, aligns prices on common dates and runs a 60-month rolling GMV-LW and $1/N$; weights at t may use data up to $t - 1$ only."* Replicate first: $1/N$ 0.76, GMV-LW 0.20 on the combined universe

Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness:** *"Draft a pre-registration: the family of tests (rules $\times$ universes $\times$ windows $\times$ costs), the error rate (FWER or FDR), the subperiods and a placebo, fixed before any result is seen."*
- **Critique:** *"Act as a referee: give the three strongest reasons my Sharpe gain is spurious and the test that would expose each."*
- **Report:** every specification tried, adjusted p -values, the power of the design, code and seeds; any specification chosen after seeing results is exploratory
- Portfolios aggregate **simple** returns and Sharpe ratios use **excess** returns: recompute one month step by step



Appendix: Further Study, Code and Data

Advanced courses on quantinar.com

- ☐  Hierarchical Clustering
- ☐  Minimum Spanning Tree
- ☐  Multivariate Statistical Analysis

Code and data

- ☐ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_04
- ☐ Risk-free rate: [Kenneth French Data Library](#)

References (1/10)

- ▣ Andrews, D.W.K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3), 817–858.
- ▣ Andrews, D.W.K., Monahan, J.C. (1992). An improved heteroskedasticity and autocorrelation consistent covariance matrix estimator. *Econometrica*, 60(4), 953–966.
- ▣ Bailey, D.H., López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
- ▣ Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300.
- ▣ Best, M.J., Grauer, R.R. (1991). On the sensitivity of mean-variance-efficient portfolios to changes in asset means: some analytical and computational results. *Review of Financial Studies*, 4(2), 315–342.
- ▣ Black, F., Litterman, R. (1992). Global portfolio optimization. *Financial Analysts Journal*, 48(5), 28–43.

References (2/10)

- Bodnar, T., Parolya, N., Schmid, W. (2018). Estimation of the global minimum variance portfolio in high dimensions. *European Journal of Operational Research*, 266(1), 371–390.
- Brandt, M.W., Santa-Clara, P., Valkanov, R. (2009). Parametric portfolio policies: exploiting characteristics in the cross-section of equity returns. *Review of Financial Studies*, 22(9), 3411–3447.
- Britten-Jones, M. (1999). The sampling error in estimates of mean-variance efficient portfolio weights. *Journal of Finance*, 54(2), 655–671.
- Chopra, V.K., Ziemba, W.T. (1993). The effect of errors in means, variances, and covariances on optimal portfolio choice. *Journal of Portfolio Management*, 19(2), 6–11.
- Choueifaty, Y., Coignard, Y. (2008). Toward maximum diversification. *Journal of Portfolio Management*, 35(1), 40–51.
- DeMiguel, V., Garlappi, L., Nogales, F.J., Uppal, R. (2009). A generalized approach to portfolio optimization: improving performance by constraining portfolio norms. *Management Science*, 55(5), 798–812.

References (3/10)

- ▣ DeMiguel, V., Garlappi, L., Uppal, R. (2009). Optimal versus naive diversification: how inefficient is the $1/N$ portfolio strategy?. *Review of Financial Studies*, 22(5), 1915–1953.
- ▣ Dragotă, M., Șerbănescu, C.I., Pele, D.T. (2008). Portfolio diversification and market share analysis for Romanian insurance companies. In *Risk Management and Value*, World Scientific, 277–298.
- ▣ Fan, J., Liao, Y., Mincheva, M. (2013). Large covariance estimation by thresholding principal orthogonal complements. *Journal of the Royal Statistical Society: Series B*, 75(4), 603–680.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Frazzini, A., Israel, R., Moskowitz, T.J. (2018). Trading costs. SSRN Working Paper 3229719.
- ▣ Frost, P.A., Savarino, J.E. (1988). For better performance: constrain portfolio weights. *Journal of Portfolio Management*, 15(1), 29–34.

References (4/10)

- ▣ Gârleanu, N., Pedersen, L.H. (2013). [Dynamic trading with predictable returns and transaction costs](#). *Journal of Finance*, 68(6), 2309–2340.
- ▣ Härdle, W.K., Simar, L. (2019). [Applied Multivariate Statistical Analysis](#), 5th ed. Springer.
- ▣ Harvey, C.R., Liu, Y. (2015). [Backtesting](#). *Journal of Portfolio Management*, 42(1), 13–28.
- ▣ He, G., Litterman, R. (2002). [The intuition behind Black-Litterman model portfolios](#). SSRN Working Paper 334304.
- ▣ Jagannathan, R., Ma, T. (2003). [Risk reduction in large portfolios: why imposing the wrong constraints helps](#). *Journal of Finance*, 58(4), 1651–1683.
- ▣ Jensen, T.I., Kelly, B., Malamud, S., Pedersen, L.H. (2026). [Machine learning and the implementable efficient frontier](#). *Review of Financial Studies*, 39(10), 3035–3078.

References (5/10)

- ▣ Jensen, T.I., Kelly, B., Pedersen, L.H. (2023). *Is there a replication crisis in finance?*. *Journal of Finance*, 78(5), 2465–2518.
- ▣ Jobson, J.D., Korkie, B.M. (1981). *Performance hypothesis testing with the Sharpe and Treynor measures*. *Journal of Finance*, 36(4), 889–908.
- ▣ Jorion, P. (1986). *Bayes-Stein estimation for portfolio analysis*. *Journal of Financial and Quantitative Analysis*, 21(3), 279–292.
- ▣ Kan, R., Wang, X., Zhou, G. (2022). *Optimal portfolio choice with estimation risk: no risk-free asset case*. *Management Science*, 68(3), 2047–2068.
- ▣ Kan, R., Zhou, G. (2007). *Optimal portfolio choice with parameter uncertainty*. *Journal of Financial and Quantitative Analysis*, 42(3), 621–656.
- ▣ Künsch, H.R. (1989). *The jackknife and the bootstrap for general stationary observations*. *Annals of Statistics*, 17(3), 1217–1241.

References (6/10)

- ▣ Laloux, L., Cizeau, P., Bouchaud, J.-P., Potters, M. (1999). Noise dressing of financial correlation matrices. *Physical Review Letters*, 83(7), 1467–1470.
- ▣ Ledoit, O., Wolf, M. (2003). Improved estimation of the covariance matrix of stock returns with an application to portfolio selection. *Journal of Empirical Finance*, 10(5), 603–621.
- ▣ Ledoit, O., Wolf, M. (2004a). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411.
- ▣ Ledoit, O., Wolf, M. (2004b). Honey, I shrunk the sample covariance matrix. *Journal of Portfolio Management*, 30(4), 110–119.
- ▣ Ledoit, O., Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.
- ▣ Ledoit, O., Wolf, M. (2017). Nonlinear shrinkage of the covariance matrix for portfolio selection: Markowitz meets Goldilocks. *Review of Financial Studies*, 30(12), 4349–4388.

References (7/10)

- ▣ Ledoit, O., Wolf, M. (2020). Analytical nonlinear shrinkage of large-dimensional covariance matrices. *Annals of Statistics*, 48(5), 3043–3065.
- ▣ Lo, A.W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52.
- ▣ López de Prado, M. (2016). Building diversified portfolios that outperform out of sample. *Journal of Portfolio Management*, 42(4), 59–69.
- ▣ Maillard, S., Roncalli, T., Teïletche, J. (2010). The properties of equally weighted risk contribution portfolios. *Journal of Portfolio Management*, 36(4), 60–70.
- ▣ Mantegna, R.N. (1999). Hierarchical structure in financial markets. *European Physical Journal B*, 11(1), 193–197.
- ▣ Marčenko, V.A., Pastur, L.A. (1967). Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4), 457–483.

References (8/10)

- ▣ Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.
- ▣ Merton, R.C. (1972). An analytic derivation of the efficient portfolio frontier. *Journal of Financial and Quantitative Analysis*, 7(4), 1851–1872.
- ▣ Merton, R.C. (1980). On estimating the expected return on the market: an exploratory investigation. *Journal of Financial Economics*, 8(4), 323–361.
- ▣ Michaud, R.O. (1989). The Markowitz optimization enigma: is 'optimized' optimal?. *Financial Analysts Journal*, 45(1), 31–42.
- ▣ Okhrin, Y., Schmid, W. (2006). Distributional properties of portfolio weights. *Journal of Econometrics*, 134(1), 235–256.
- ▣ Pele, D.T. (2023). Robust Markowitz for crypto-assets. Quantinar course, Bucharest University of Economic Studies.

References (9/10)

- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). *Are cryptos becoming alternative assets?*. *The European Journal of Finance*, 29(10), 1064–1105.
- ▣ Petukhina, A., Trimborn, S., Härdle, W.K., Elendner, H. (2021). *Investing with cryptocurrencies – evaluating their potential for portfolio allocation strategies*. *Quantitative Finance*, 21(11), 1825–1853.
- ▣ Politis, D.N., Romano, J.P. (1994). *The stationary bootstrap*. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- ▣ Sharpe, W.F. (1966). *Mutual fund performance*. *Journal of Business*, 39(S1), 119–138.
- ▣ Theil, H., Goldberger, A.S. (1961). *On pure and mixed statistical estimation in economics*. *International Economic Review*, 2(1), 65–78.
- ▣ Tobin, J. (1958). *Liquidity preference as behavior towards risk*. *Review of Economic Studies*, 25(2), 65–86.

References (10/10)

- ▣ Tu, J., Zhou, G. (2011). Markowitz meets Talmud: a combination of sophisticated and naive diversification strategies. *Journal of Financial Economics*, 99(1), 204–215.
- ▣ Data: Kenneth R. French Data Library, mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.